

Response to Referee 2

Manuscript JFM-2026-1781

Strain-coupled one-dimensional turbulence for rapid distortion

Sharma & Kerstein

We thank the referee for the report and for the specific reading recommendations, in particular the pointer to Sagaut & Cambon (2018), which has been adopted as the rapid-distortion reference spine. The manuscript has been substantially rewritten: it is reorganised into three declared parts around a single take-home message, Section 2 is compressed and its two halves connected, redundant closure discussion is consolidated, an independent DNS benchmark has been added, and the leading-edge application flagged in the old conclusions is now carried out. Section, figure and table numbers below refer to the revised manuscript.

1. Length, readability, and the absence of a clear take-home message.

“Under its present form, this very long (42 pages) manuscript is very difficult to read, with no clear take-home message(s) on its relevant and original results. Often, similar discussions and models took place in different sections (and plenty of sub- and subsubsections), with unuseful redundancy.”

Response. The paper is restructured into three explicitly announced parts (§1, closing paragraphs): the model and what a single line can know (§§2–5); the measured limits and the relaxation mechanism (§6); and the consequence for leading-edge noise (§7). The abstract and the introduction now state the take-home message in one line—that the model evolves the strained spectrum, carrying rapid distortion and nonlinear relaxation together, and that the difference this makes to a downstream (leading-edge-noise) prediction is a term too large to neglect. The six contributions are listed once, as an itemised list in §1. Redundant discussion has been removed by moving the CMK comparison, the full intervention catalogue, the implementation-verification detail and the leading-edge-chain steps to electronic supplementary material (§§S1–S5) and to Appendices A–B, leaving each topic in a single place in the main text.

2. Section 2, and the disconnected subsections 2.1 and 2.2.

“Section 2 illustrates what is the most difficult to understand ... with almost disconnected sections 2.1 and 2.2. ... Since the application to flows with mean shear has been published to JFM in 2001 by Kerstein et al., I wonder if a clear review of the JFM paper should be better than the ms. subsection 2.1?”

Response. Section 2 has been reorganised around a single organising principle—the separation of the distortion physics by structure (production and a spectrally closed rapid operator as continuous forcing; the eddy events as the turbulence-driven relaxation)—which is stated at the outset and carried through to the summary (§2.6). Following the referee’s suggestion, the

standard-ODT subsection (§ 2.1) is now a compact review keyed to Kerstein (1999) and Kerstein *et al.* (2001) rather than a self-contained re-derivation. The link the referee found missing is made explicit: § 2.2 derives the strain-coupled line equation and its structural admissibility directly from the ODT kernel of § 2.1, and § 2.3 connects the Reynolds-stress/RDT description to that line equation through the rapid/slow pressure split, so 2.1 and 2.2 are now two stages of one construction rather than two reviews.

3. Sagaut & Cambon (2018) as the reference set; RDT solution machinery.

“all details for solving RDT equations (2.14) by a tensorial kernel are gathered in the monograph by Sagaut & Cambon ... a new edition 2018 ... is of particular interest with its more complete chapter 8 on turbulence with strain ... often more recent and more relevant than the ones of the ms.”

Response. Sagaut & Cambon (2018) is now cited as the modern systematic reference for rapid distortion in the introduction and is used as the reference for the RDT solution machinery where the wavevector-ensemble integration is introduced (§§ 1, 2.3). We have taken the referee’s point that the exact rapid pressure–strain and its spectral integration are standard results and now present them as such, attributed to the classical sources and the monograph, and separated from what is new here (the single-line reduction, its bounds and the measured axisymmetry hypothesis in § 5).

4. What are the objectives and achievements; the model description.

“What are really the objectives and the achievements of the study remain very difficult to size. ... Near the end of the paper, before Eq. (5.4) a useful possible description of the model is given: ODT evolves a velocity vector $u'(y)$ on a single line aligned with the resolved direction ...”

Response. The objectives and achievements are now stated up front: the abstract gives the aim and the outcome, and § 1 lists the six results with the section where each is established and, in each case, the reference data it is measured against. The concise description of the model the referee found useful late in the paper has been moved forward: the line and its observables are defined where the single-line question is set up (§ 5.2, equation (5.3)), and the operational picture—ODT evolving $u'(y)$ on the resolved axis, with production and the rapid operator as continuous forcing and the eddy events as relaxation—is now given in the formulation and its summary (§§ 2.2, 2.6).

5. The closure appears in many places.

“The related problem of closure appears in different sections, subsections, subsubsections ... p. 37, 5.3. Eqs. (5.7)-(5.10). Perhaps the sole original result for the closure of $\Pi_{ij}^{(r)}$?”

Response. The closure material is consolidated. The exact rapid term and the closure problem are stated once in the formulation (§ 2.4.1); the isotropic moment closure actually used in the simulations is given once (§ 2.4.2); and the single-line treatment—which the referee identifies as the original contribution—is collected in Section 5, where the collapsed spectral form (now equations (5.6)–(5.9)) and its kernels appear, with the derivation in Appendix A.3. Section 5.7 then shows explicitly that this spectral form reduces to the moment closure used earlier, so

the two appearances of “closure” are the same object at two levels of resolution rather than separate treatments. We agree with the referee’s reading that Section 5 is the original result, and the paper is now built around it.

6. Validation, and the reliance on the 1985 Lee & Reynolds DNS.

“Validation of (several versions of) the model is not clear. Why the author favours the old DNS by Lee and Reynolds (1985)? ... None of the 15 figures of the ms. gives a convincing validation of a new model, capable of incorporating RDT and nonlinearity, vs. numerical or experimental results.”

Response. Validation has been strengthened and its versions delimited. Lee & Reynolds (1985) is retained because it resolves a Reynolds number a single line can match and gives a graded moment-level target, but it is now one condensed subsection (§ 3.4) and its figures (4–6) carry the independent DNS of Zusi & Perot (2013, 2014) as an overlay, which agree where they overlap. The convincing spectral validation the referee asks for is new: figure 11 (with table 1) is a three-way comparison of the strain-induced spectral anisotropy against our own 128^3 strained-box DNS and its exact-RDT companion at matched projection; figure 12 quantifies the distance from linear theory versus strain rapidity; figure 16 extends this to a five-way comparison across three rapidities; and figure 19 tests the assembled leading-edge-noise chain against four published experiments. The several “versions” are labelled explicitly (category.variant, table 4 and supplementary § S2) so that which model produced which figure is unambiguous.

7. Introduction references far from the model; shock-turbulence.

“The introduction incorporates many references of studies of sound radiation, that are very far from the model developped in the ms. Idem with the shock-turbulence interaction ...”

Response. The introduction has been rewritten and trimmed. The far-field sound-radiation citations not connected to the distortion problem have been removed, and the shock-turbulence material has been dropped; the retained acoustics references are confined to the leading-edge-noise framework that consumes the model output. The introduction now moves from the distortion problem to the specific gap the model fills, without the surrounding survey.

8. The conclusions open an application that is not carried out.

“The last paragraph of the conclusions seems to open the way to an application to complex flow: ... the distorted, anisotropic upwash spectrum at the leading-edge plane ... I think that such an objective merits to be supported, but this is not really done in the present ms.”

Response. That application is now carried out in the body of the paper rather than deferred. Section 7 supplies the computed distorted upwash spectrum to Amiet’s leading-edge-noise theory, both as a geometry-independent kernel ratio (figure 17) and as an absolute far-field prediction at the reference operating point (figure 18), quantifies the 5–20 dB difference from the frozen-spectrum practice, and compares the assembled prediction with four published experiments (figure 19), agreeing within 1–2 dB rms where the facilities are cleanest. The conclusions now report this result rather than proposing it.